

Autonomous Tennis Analysis

Team B

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Problem

What did we set out to achieve?

The Goal:

To extract meaningful tennis match analytics such as player position, movement patterns and ball trajectory and visualise these statistics in real time. We aim to use classical computer vision techniques and benchmark our pipeline against state-of-the-art ML models

Why is it challenging?

- Variable camera angles, lighting conditions and court surfaces
- Small, fast-moving ball subject to motion blur
- Cluttered broadcast footage with auxiliary shots of crowd or irregular angles

Why Computer Vision?

- Enables automated and rapid analysis at scale without manual annotation. Could enable real time coaching insight and performance analytics.

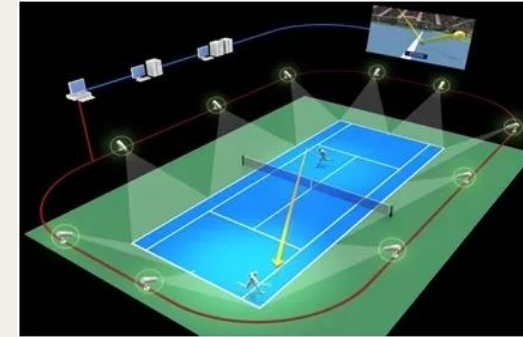


Literature Review

Approaches in Sport Video Analysis

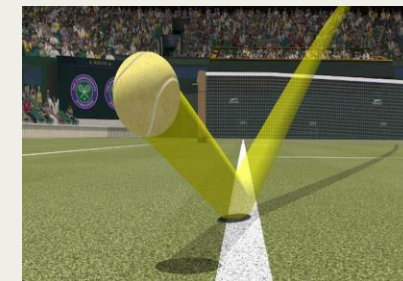
Hawk-Eye:

- Uses 10-12 high-speed cameras
- Tracks ball from multiple angles → builds a 3D trajectory model.
- Accuracy of 1-2mm for line calls
- ~ 100,000 USD per court hardware costs, setup taking 3+ days



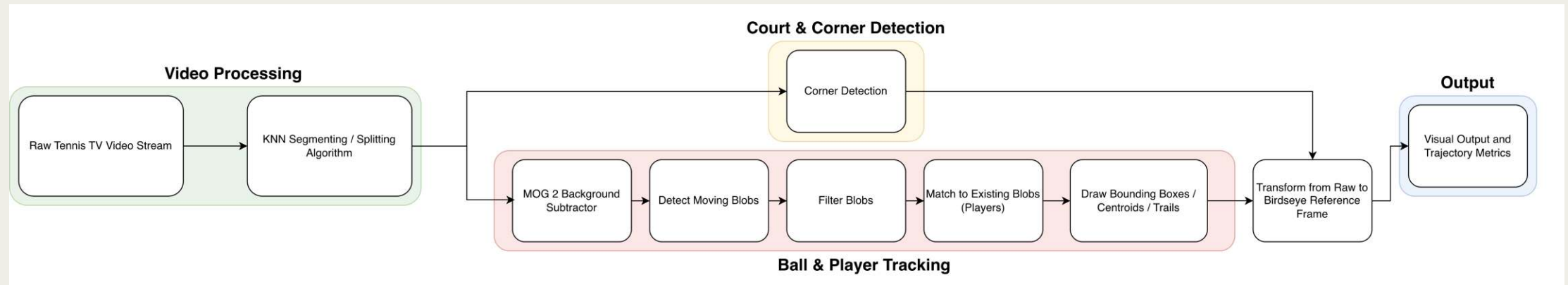
Existing Literature Approaches:

- Hybrid models combining Hough & Deep Learning for court detection
- Geometric approaches enabling multi-view court calibration and tracking
- Real time YOLO models handling players efficiently, R-CNN methods for pose based tracking with differing trade-offs between the two.
- Physics based bounce/spin/skid/compression models to improve ball tracking



Overall System Architecture

Our Pipeline: from TV broadcast to Birds Eye tracking view



Method

Intelligent Frame Filtering & Court Localisation

Frame Classification

- **Lightweight feature extraction** (white line mask + line segment detector + orientation histogram) **KNN classifier** network designed to classify raw tennis TV footage as *gameplay* or *auxiliary shots*

Gameplay

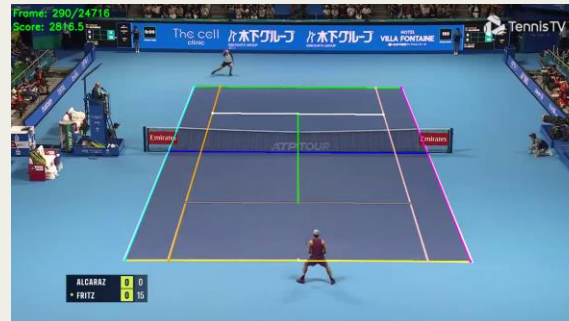


Auxiliary



Court Corner Detection

- Robustly determine the corners of each tennis court in a new sequence to establish a ground truth reference point for warping and metric plotting
- Canny/ Hough line detection parameters were unstable across differing lighting and court conditions
- Incorrect or missing corner points
- White line segmentation proved stable in artificial courts
- Diverse dataset made it difficult to tune parameters robustly



Method

Dynamic Ball & Player Tracking → Mapping to Birds Eye

Player Tracking

- Background subtraction using Mixture of Gaussians (MOG2)
- Robust method due to camera stability
- Clustered Blob & Contour detection
- Geometric constraints enforced for increased accuracy

Ball Tracking

- Difficult to track due to small size, motion blur and ball colour closely matching background hues
- Hard-coded colour-based segmentation was ruled out
- Uses similar principles as player tracking

Warping to Birds Eye

1. Court Corner detection
2. Define real world court dimensions
 1. Convert pixels to metres scale factor
3. Compute homography matrix that maps points from oblique view to bird's eye view.
4. Warp player and ball centroids from camera space to court space.

Calculate Metrics

- Convert distance travelled in pixels – multiplying by scale factor.

Visualisation

- Create top-down court visualisation with player and ball positions

Experimental Setup

Evaluation Strategy against SOTA



Tennis Dataset

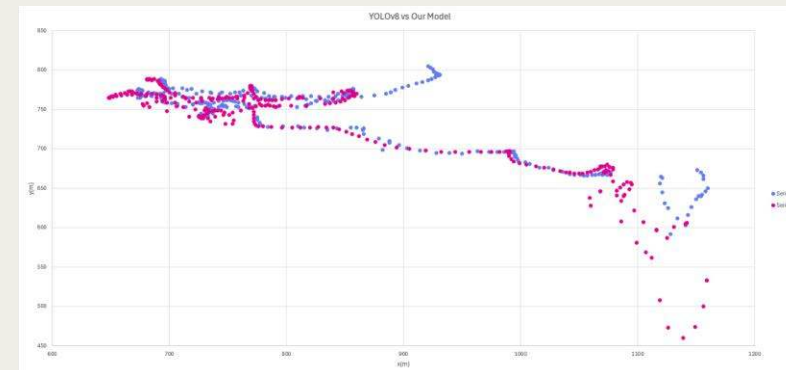
Evaluation Metrics

TP	FN
FP	TN

Frame Detector &
Court Points
Confusion Matrix



Benchmark against
SOTA detectors



Detected Player & Ball Centroid comparison
against ground truth

DEMO

Conclusion and Future Work

Reflections & Next Steps

Limitations

- Ball tracking
- Court detection
- Broadcast image frame quality
- Real time processing capabilities and computational overhead.

Future Improvements

1. Implement more sophisticated ball detection to improve scene robustness
2. Train a network to classify and label shot types.
3. Implement a physics network tailored to tennis for trajectory prediction and to improve Kalman motion filter
4. Calculate advanced player/game metrics
5. Test real time feasibility

APPENDIX

Frame Classification



“Net proxy” (horizontal band energy)

Measures vertical edge strength across the center band of the frame.

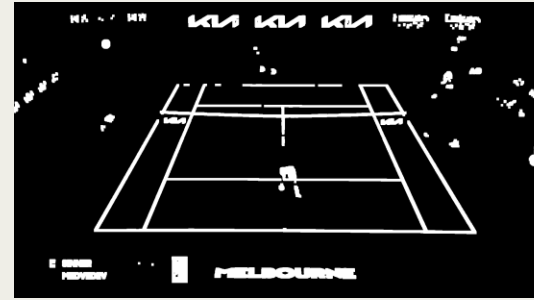
The net creates strong horizontal edges, producing large vertical gradients.



Line segment detection

Uses the Line Segment Detector (LSD) to find straight lines in the image.

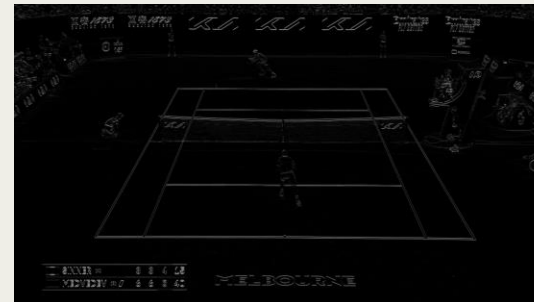
Valid broadcast views contain many long, consistent lines compared to close-ups.



Court line fraction

Applies a white-line mask to detect bright, low-saturation regions (court markings).

Frames with a higher fraction of white lines are more likely to show the full court.



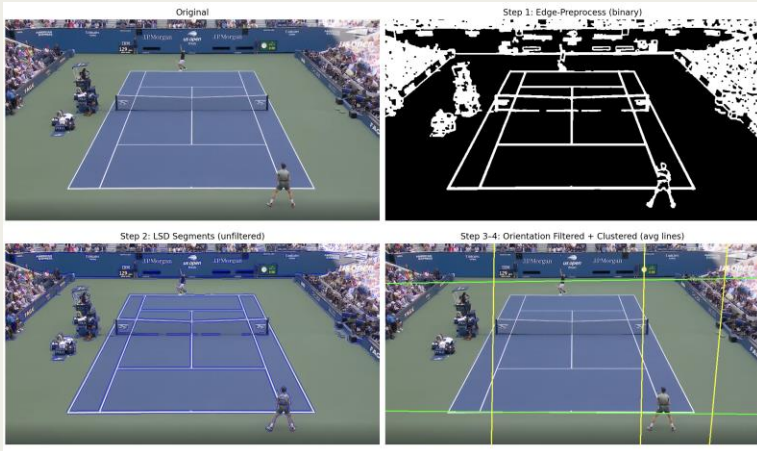
Orientation structure

Analyses the distribution of line angles in a histogram.

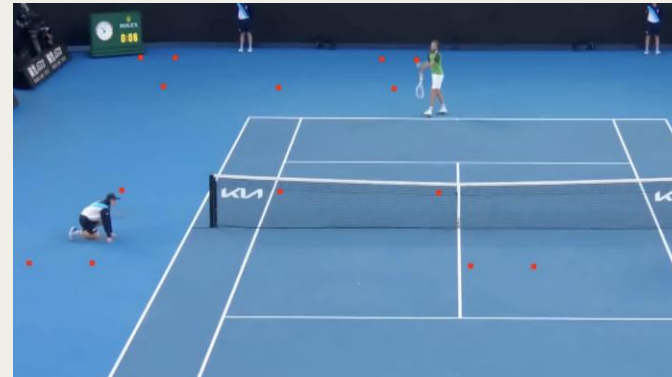
Valid court views show ordered orientations (horizontal, vertical, diagonal), while crowd shots exhibit random edge directions.

Roadblocks

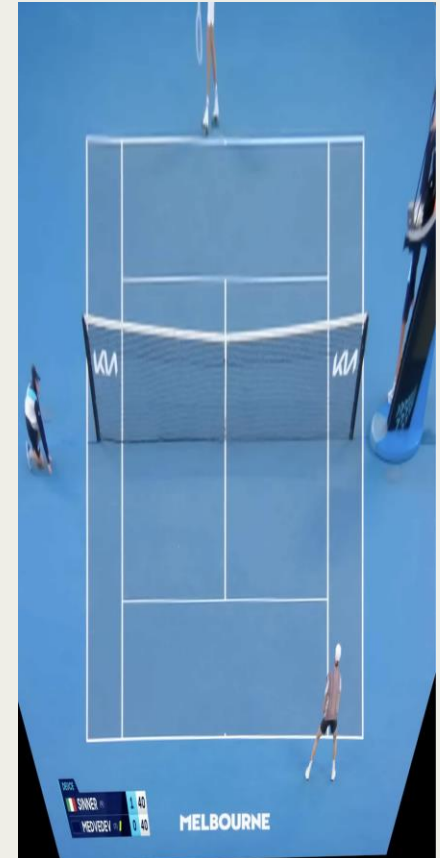
Corner Detection



Classical Vision Methods



Untrained model & Image sizing mismatch



No source pixels

Confusion Matrix

Corner Detection

	Predicted Positive	Predicted Negative
Actual Positive	TP = 29,659	FN = 50
Actual Negative	FP = 1,146	TN = 99

Metric	Results
Precision	0.963
Accuracy	0.962
Median Distance	1.837

No source pixels